# Statistical and Machine Learning

Clustering analysis

## Instructions

All questions should first be discussed in pairs or small groups. Afterwards, we shall discuss them together. Solutions will be distributed after each session.

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**Question 1: Clustering fundamentals**

(a) In your own words, what is the difference between **supervised** and **unsupervised** learning? Where does clustering fit?

(b) A dataset has $N = 100$ observations in $\mathbb{R}^5$. You are given an assignment function $a(\cdot)$ that maps each observation to one of $K = 3$ clusters. State the two conditions that a good assignment function should satisfy in terms of within-cluster and between-cluster distances.

(c) Give **two** examples of dissimilarity measures $D_{ij}$ that can be used for quantitative variables.

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**Question 2: Purity and Rand Index**

Consider four clusters with $N = 20$ objects and true class labels A, B, C:

| Cluster | Contents | $N_i$ |
|---|---|---|
| 1 | A A A A B | 5 |
| 2 | B B B A | 4 |
| 3 | C C C C C B | 6 |
| 4 | A A C C C | 5 |

(a) Compute the purity $p_i$ of each cluster, and the overall purity of the clustering.

(b) Explain in one sentence the key **limitation** of purity as an evaluation metric.

(c) For the Rand index, define $TP$, $TN$, $FP$, and $FN$ in terms of pairs of objects. What familiar classification metric does the Rand index correspond to?

(d) Suppose a new clustering has $TP = 30$, $FP = 10$, $FN = 15$, $TN = 45$. Compute the Rand index $R$.

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**Question 3: Hierarchical Agglomerative Clustering**

(a) Describe in plain language what the **dendrogram** produced by HAC represents. What does the height of a node correspond to?

(b) You are given five points and told that single-link clustering is used with city block distance. The first merge occurs between points $(2, 4)$. Explain what this means about the distances in the data.

(c) Complete the table below comparing the three linkage criteria:

| Linkage | Formula for $d(G, H)$ | Properties* | Scale invariant? |
|---|---|---|---|
| Single | | | |
| Complete | | | |
| Average | | | |

- By Properties, we refer to the shape of clusters (i.e., elongated, compact, well-separated, etc) as well as to the properties of the method (i.e., sensitive to outliers)

(d) Give **one advantage** and **one disadvantage** of HAC compared to K-means.

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**Question 4: K-means clustering**

(a) Write down the **distortion** objective $J(\mathbf{M}, \mathbf{Z})$ minimised by K-means. What are $\mathbf{M}$ and $\mathbf{Z}$?

(b) Describe the two steps of the K-means algorithm (assignment step and update step). Why is the algorithm guaranteed to converge?

(c) Why does the distortion $J$ **not** provide a valid criterion for choosing the number of clusters $K$? What does the **elbow method** suggest instead, and why is it sometimes unreliable?

(d) The **silhouette coefficient** for point $i$ is $sc(i) = \frac{b_i - a_i}{\max(a_i, b_i)}$. Interpret the three boundary cases $sc(i) \approx +1$, $sc(i) \approx 0$, and $sc(i) \approx -1$.

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**Question 5: Mixture Models and the EM Algorithm**

(a) Write down the density of a **Gaussian mixture model (GMM)** with $K$ components. Identify all parameters and state the constraint on the mixing weights $\pi_k$.

(b) Define the **responsibility** $r_{nk}$ in a GMM. What does it represent, and why is it called a "soft" assignment?

(c) Briefly describe what happens in the **E step** and the **M step** of the EM algorithm when fitting a GMM. What quantity is guaranteed to be non-decreasing at each iteration?

(d) State the **two approximations** that reduce EM for a GMM to the K-means algorithm.

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**Question 6: Spectral Clustering**

(a) Why do K-means and GMMs struggle with **non-convex** clusters? Give an example of a dataset shape where spectral clustering would be expected to outperform K-means.

(b) Define the **graph Laplacian L** in terms of the weight matrix $\mathbf{W}$ and the degree matrix $\mathbf{D}$. What is the degree $d_i$ of node $i$?

(c) State Theorem on Graph Laplacian about the eigenspace of $\mathbf{L}$ with eigenvalue 0. What does this theorem tell us about the relationship between eigenvectors and cluster structure?

(d) List the steps of the **Ng–Jordan–Weiss** spectral clustering algorithm.

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**Question 7: Comparing clustering methods**

Fill in the table below. For each property, mark which methods satisfy it with a tick, cross, or brief note.

| Property | HAC | K-means | GMM + EM | Spectral |
|---|---|---|---|---|
| Requires $K$ in advance | | | | |
| Has a well-defined objective | | | | |
| Produces soft (probabilistic) assignments | | | | |
| Handles non-spherical clusters | | | | |
| Deterministic (no random initialisation) | | | | |
| Scales well to large $N$ | | | | |

In 2–3 sentences, justify your choice for the row "Handles non-spherical clusters".